

Spring 2014

Problem 1 (Sp14). Find the sum of the series

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \cdots$$

Problem 2 (Sp14). Prove or disprove that there is a sequence $\{f_n\}$ of continuous functions on $\mathbb{R}$ such that for any rational x the sequence $(f_n(x))$ is bounded but the sequence $\left(f_n\left(x + \sqrt{2}\right)\right)$ is unbounded.

Problem 3 (Sp14). Find a real number c so that

$$\left| c - \int_{-1/2}^{1/2} \frac{\exp(x) - 1}{x} dx \right| < 0.01$$

Problem 4 (Sp14). Let f be analytic on the closed unit disc, and assume that $|f(z)| \leq 1$ for all z 's in this set. Suppose also that $f(1/2) = f(i/2) = 0$. Prove that $|f(0)| \leq 1/4$.

Problem 5 (Fa79, Su81, Sp14). Suppose f and g are entire functions with $|f(z)| \leq |g(z)|$ for all z . Prove that $f(z) = cg(z)$ for some constant c .

Problem 6 (Sp14). Let R be a finite ring with identity of characteristic p . For S a subring of R , not necessarily containing an identity element, S is a vector space over $\mathbb{F}_p$. For $a \in S$ let $T_a^S : S \rightarrow S$ be the linear map $T_a^S(x) = ax$ and $T_a^R : R \rightarrow R$ be the linear map $T_a^R(x) = ax$

1. Show that if $1 \in S$ then the minimal polynomial of T_a^S = the minimal polynomial of T_a^R .
2. Give an example of p, R, S, a showing that the hypothesis $1 \in S$ cannot be dropped in Part 1.

Problem 7 (Sp14). Let $\mathbb{F}_q$ be a finite field with q elements. A *complete flag* in the vector space $\mathbb{F}_q^n$ is a nested sequence of linear subspaces $V^1 \subset V^2 \subset \cdots \subset V^{n-1}$ of dimensions $1, 2, \dots, n-1$, respectively. Determine the number of complete flags in $\mathbb{F}_q^n$ as a function of n and q and find the limit of $f_n(q)$ as q tends to 1.

Problem 8 (Sp14). Let G be a group of order 48. Show that G contains a normal subgroup of order 16 or 8.

Problem 9 (Sp14). Let A be a finite abelian group and let $R = \text{End}_A$ be the ring of homomorphisms from A to A . Show there is a subring S of R such that A and S are isomorphic as abelian groups.

Problem 10 (Sp14). Let the continuously differentiable function $f : [0, 1]^2 \rightarrow \mathbb{R}$ on the unit square be given by the distance to a fixed point outside the square. Show that there is no point (x_0, y_0) in the square such that the gradient of the function f at this point is equal to the average value of the gradient of the function f . In other words the obvious analogue of the mean value theorem for functions of two variables is false.

Problem 11 (Fa05, Sp14). Let $K \subset \mathbb{R}^n$ be closed, convex, and nonempty. Show that for every $x \in \mathbb{R}^n$, there exists $y \in K$ that uniquely minimizes the Euclidean distance to x , i.e., $\|x - y\| < \|x - z\|$ for all $z \in K \setminus \{y\}$.

Problem 12 (Sp14). Prove that there exists a constant C such that for every polynomial P of degree 2014

$$P(0) \leq C \int_0^1 |P(x)| \, dx$$

Problem 13 (Su83, Sp14). Compute the area of the image of the unit disc $\mathbb{D} = \{z \mid |z| < 1\}$ under the map $f(z) = z + z^2/2$.

Problem 14 (Fa02, Sp14). Evaluate $\int_{-\infty}^{\infty} \frac{\cos x}{(1 + x^2)^3} \, dx$.

Problem 15 (Su81, Sp14). Let $\text{SO}_3(\mathbb{R})$ denote the group of orthogonal transformations of $\mathbb{R}^3$ of determinant 1. Let $Q \subset \text{SO}_3(\mathbb{R})$ be the subset of symmetric transformations $\neq I$. Let P^2 denote the space of lines through the origin in $\mathbb{R}^3$.

1. Show that P^2 and $\text{SO}_3(\mathbb{R})$ are compact metric spaces, in their usual topologies.
2. Show that P^2 and Q are homeomorphic.

Problem 16 (Sp88, Sp14). Compute A^{10} for the matrix

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{pmatrix}$$

Problem 17 (Sp14). Let $B = C^{-1}AC$, where A and C are $n \times n$ matrices with integer entries, such that $\det A = 1$, and $\det C \neq 0$. Prove that there exists a positive integer m such that all entries of B^m are integers.

Problem 18 (Fa99, Sp14). Let G be a finite group acting on a finite set X with a single orbit. For an element $g \in G$ let $\text{Fix}_g(X)$ denote the set $\{x \in X \mid g(x) = x\}$.

1. Show that $|G| = \sum_{g \in G} |\text{Fix}_g(X)|$
2. Show that if X has more than one point then there exists an element $g \in G$ fixing no points of X .